

L.J Institute of Engineering and Technology, Ahmedabad.
INS Practice book (SEM-VI-2026 CE,IT,CSD,AIML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
UNIT 1 INTRODUCTION								
Topic Name:- Security Attacks, Security services and Mechanism								
1	1	_____ attack attempts to learn or make use of information from the system but does not affect system resources.	A	1	Passive	Active	Replay	Masquerade
2	1	In order to ensure the security of the Message/information, we need to _____ the data.	B	1	Decrypt	Encrypt	Delete	Delete and Encrypt
3	1	What do you mean by Brute-force attack?	A	1	The attacker tries every possible key on a piece of ciphertext until plaintext is obtained.	Transforming an intelligible message into one that is unintelligible.	Retransforming that message back to its original form.	Types of operations used for transforming plaintext to ciphertext.
4	1	_____ attack attempts to alter system resources or affect their operation.	A	1	Active	Passive	Replay	Masquerade
5	1	_____ Specifies that only the sender and the intended recipient should be able to access the contents of a message.	C	1	Integrity	Antivirus	Confidentiality	proof of identities
6	1	_____ is the practice and precautions taken to protect valuable information from unauthorized access, recording, disclosure or destruction.	A	1	Information Security	Database Security	Network Security	Physical Security
7	1	_____ is a System Design to prevent unauthorized access to or from a Private network.	B	1	Unauthorized Attacks	Firewalls	Virus Attacks	Fire Attacks
8	1	What are the components of information security?	D	1	Confidentiality	Integrity	Availability	All of these
9	1	Authentication means _____.	A	1	Verification of user's identification	Verification of the data	Verification of the Sender only	Verification of the Receiver only
10	1	Encryption and decryption provide secrecy, or confidentiality, but not _____.	B	1	Authentication	Integrity	Privacy	Non Repudiation
11	1	_____ is the assurance that someone cannot deny something.	C	1	Authentication	Data integrity	Nonrepudiation	Data confidentiality
12	1	_____ Determines who should be able to access what.	D	1	Authentication	Data Confidentiality	Data Integrit	Access Control
13	1	The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts is called _____.	A	1	Traffic Padding	Routing Control	Notarization	Digital Signature
14	1	_____ is a process that is designed to detect, prevent, or recover from a security attack.	D	1	Security Attack	Security Service	Traffic analysis	Security Mechanism
15	1	Provides integrity of all user data on a connection and detects any modification, insertion, deletion, or replay of any data with recovery attempted is called _____.	A	1	Connection Integrity with Recovery	Connection Integrity without Recovery	Selective-Field Connection Integrity	Connectionless Integrity
TOPIC NAME:- Substitution and Transposition Techniques								
16	1	Construct a playfair matrix with the key "occurrence". Generate the cipher text for the plaintext "Tall trees"	A	1	PF IZ TZ EO RT	LC YO GW BE TR	PI TU UE FL GX	PM FL PG MT HK
17	1	Encrypt the Plaintext message "EXAM" with Key "HILL" using the Hill cipher.	C	1	FMIV	ELSM	ELSC	FIVM
18	1	_____ is a relationship between characters in the plaintext to a character in the ciphertext is one-to-many.	C	1	Monoalphabetic Cipher	Playfair cipher	Polyalphabetic cipher	One Time Pad
19	1	Plaintext is written down as a sequence of diagonals and then read off as a sequence of rows is called _____.	C	1	Rows/Columns Transposition	One Time Pad	Rail fence technique	Vernam Cipher
20	1	_____ replace each letter of the alphabet with the letter standing 3 places further down the alphabet.	D	1	Monoalphabetic Substitution Cipher	Playfair Cipher	Hill Cipher	Caesar cipher

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21	1	Construct a playfair Matrix with the key "moonmission" and encrypt the message "greet".		3				
22	1	Construct a playfair matrix with the key "occurrence". Generate the cipher text for the plaintext "Tall trees".		3				
23	1	Construct 5 X 5 playfair matrix for the keyword "OCCURANCE".		2				
24	1	Construct a Playfair matrix with the key "engineering". And encrypt the message "test this process".		3				
25	1	Let the keyword in playfair cipher is "keyword". Encrypt a message "come to the window" using playfair cipher.		4				
26	1	Encrypt the following message using playfair cipher. Message: COMSEC means communications security Keyword: Galois.		3				
27	1	Using playfair cipher encrypt the plaintext "Why, don't you?". Use the key "keyword".		3				
28	1	Use playfair algorithm with key "monarchy" and encrypt the text "jazz".		2				
29	1	Explain playfair cipher substitution technique in detail. Find out cipher text for the following given key and plaintext. Key = ENGINEERING Plaintext=COMPUTER		3				
30	1	Explain Playfair Cipher in detail. Find out cipher text for the following given plain text and key. Key = GOVERNMENT Plain text = PLAYFAIR		3				
31	1	Encrypt the message "meet me at the usual place " using the Hill cipher using key $\begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$		2				
32	1	Use Hill cipher to encrypt the text DEF. The key to be used is $\begin{bmatrix} 2 & 4 & 5 \\ 9 & 2 & 1 \\ 3 & 8 & 7 \end{bmatrix}$		3				
33	1	Given key K= $\begin{bmatrix} 17 & 17 & 05 \\ 21 & 18 & 21 \\ 02 & 02 & 19 \end{bmatrix}$ and plaintext ="ney". Find out the ciphertext applying Hill Cipher.		3				
34	1	Encrypt Message "Secure" using Hill Cipher with key $\begin{bmatrix} 17 & 10 \\ 23 & 19 \end{bmatrix}$		3				
35	1	Encrypt the given message using playfair cipher. Message : GOOD MORNING Key : GTU EXAMS		3				
36	1	Encrypt the Message "Surgical Strike" with key "GUJAR" using PLAYFAIR technique.		2				

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37	1	(1) Encrypt the text “trust” using Caesar Cipher with key=25 (2) Compare the strength of mono-alphabetic ciphers to poly-alphabetic ciphers.		2				
38	1	Encrypt the message “GTU Examination” using the Hill cipher algorithm with the key matrix $\begin{bmatrix} 5 & 17 \\ 4 & 15 \end{bmatrix}$ Show your calculations and the result.		3				
39	1	Perform encryption in Playfair Cipher algorithm with plain text as“INFORMATION AND NETWORK SECURITY”, Keyword is“MONARCHY”. (Note: 1.Put j and i both combine as a single field in 5*5matrix).		3				
40	1	Construct a Playfair matrix with the key “engineering”. And encrypt the message “impossible”.		2				
41	1	Encrypt the message "SUPERSPECIALITY HOSPITAL" by 9 place down by ceaser cipher.		2				
42	1	Find cipher text for plain text ‘GTUINSEXAM’ using ‘STUDY’ as key by Playfair cipher method.		2				
UNIT -2 Basics of Cryptography								
TOPIC NAME:- Characteristics of Cryptography and Cryptoanalysis								
43	2	Cryptanalysis is used _____	A	1	to find some insecurity in a cryptographic scheme	to increase the speed	to encrypt the data	to make new ciphers
44	2	In cryptography, what is cipher?	A	1	algorithm for performing encryption and decryption	encrypted message	both algorithm for performing encryption and decryption and encrypted message	decrypted message

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45	2	What do you understand by cryptography and cryptoanalysis ?		2				
46	2	What are the defining characteristics of cryptography?		2				
TOPIC NAME:- Block cipher modes of operation								
47	2	What about encrypting plaintext longer than b bits in Block Cipher?	B	1	Operates on fixed length b-bit input to produce b-bit ciphertext.	Break plaintext into b-bit blocks (padding if necessary) and apply cipher on each block.	Plaintext handled one block at a time and each block of plaintext is encrypted using the same key.	It operates on different length.
48	2	The contents of the shift register are shifted left by s bits, and C1 is placed in the rightmost (least significant) s bits of the shift register. In which Block Cipher Mode Operation Above statement is used.	B	1	Cipher Block Chaining (CBC)	Cipher Feedback (CFB)	Output Feedback (OFB)	Counter (CTR)
49	2	In OFB Transmission errors do not propagate: only the current ciphertext is affected, since keys are generated "locally".	A	1	TRUE	FALSE	May be	Can't say
50	2	Which mode requires the implementation of only the encryption algorithm?	C	1	ECB	CBC	CTR	OFB
51	2	A _____ is one that encrypts a data stream one bit or one byte at a time	A	1	Stream cipher	Block cipher	Feistel Cipher	Stream cipher and Block cipher
52	2	Which of the following statements are true? i) In the CBC mode, the plaintext block is XORed with previous ciphertext block before encryption ii) The CTR mode does required counter iii) The last block in the CBC mode uses an Initialization Vector iv) In CBC mode repetitions in plaintext do not show up in ciphertext	C	1	iii only	ii and iv	I, ii and iv	i only
53	2	Which of the following modes does not implement chaining or "dependency on previous stage computations"?	A	1	CTR, ECB	CTR, CFB	CFB, OFB	ECB, OFB
54	2	Which one of the following algorithm is not used in asymmetric-key cryptography?	C	1	RSA algorithm	Diffie-Hellman algorithm	Electronic Code Book algorithm	DSA algorithm
55	2	What is your understanding of Cipher Feedback (CFB) block cipher modes of operation.		2				
56	2	What is the limitation of Electronic Codebook Mode (ECB)? How it is overcome by Cipher Block Chaining (CBC) mode?		2				
57	2	What do you understand by Cipher Block Chaining (CBC) and Electronic Code Book (ECB) block cipher modes of operation.		3				
UNIT 3 - Block Cipher								
TOPIC NAME:- Block Cipher,AES, modes of AES								
58	3	For the AES-128 algorithm there are similar rounds _____ and _____ round is different.	B	1	2 pair of 5 similar rounds, every alternate	9 , the last	8 , the first and last	10 , no
59	3	AES stands for _____.	A	1	Advanced Encryption Standard	Advanced Encryption System	Advanced Encryption Suggestion	None of above
60	3	_____ developed AES.	C	1	MSIT	NSIT	NIST	MIST
61	3	NIST stands for	B	1	National Institute of Security and Technology	National Institute of Standards and Technology	National Institute of Standards and Terminology	Northern Institute of Standards and Terminology

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62	3	AES was created in _____ and was released in _____.	A	1	1997 & 2001	2001 & 1997	1998 & 2001	2001 & 1998
63	3	AES uses a _____ bit block size and a key size of _____ bits.	D	1	128; 128 or 256	64; 128 or 192	256; 128, 192, or 256	128; 128, 192, or 256
64	3	What is the expanded key size of AES-192?	D	1	32 words	64 words	44 words	52 words
65	3	Aes works on ____*____ mtrix.	B	1	3*3	4*4	5*5	6*6
66	3	The 4x4 byte matrices in the AES algorithm are called	A	1	States	Words	Transitions	Permutations
67	3	Which predefined matrix is used in mix column of AES.	A	1	$\begin{bmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 3 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 2 & 1 & 2 \end{bmatrix}$
68	3	How many AES rounds are required for 256-bit key size?	A	1	14	12	4	16
69	3	In AES for howmany key size 10 rounds are required?	A	1	128	192	162	256
70	3	How many rounds does the AES-192 perform?	B	1	10	12	4	16
71	3	AES the 4x4 bytes matrix key is transformed into a keys of size _____.	C	1	32 words	64 words	44 words	54 words
72	3	How to generate the Rotword?	A	1	It performs an one byte circular left shift on a word.	It performs a two byte circular left shift on a word.	It performs a Three byte circular left shift on a Word.	It performs a one byte circular left shift and a three byte circular left shift on a word
73	3	What is the output of following AddRoundKey in AES Algorithm? 32 + (XOR) 2b= _____	D	1	e9	9a	a0	19
74	3	Which steps are used in AES 1) Substitute Bytes 2) Shift column 3) Mix row 4) Add Round Key	C	1	only 1	1 & 2	1 & 4	All
75	3	The input 128 bit block, this block is arranged in the form of 4 X 4 square matrix of bytes. This block is copied into the	A	1	state array	Array	Table	Block

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76	3	The AES key expansion algorithm takes as input a (16-byte) key and produces a linear array of _____.	D	1	Six word and 40 words	Four word and 39 words	Five word and 42 words	Four word and 44 words																																																																																
77	3	what will be the output after performing the third operation shift row on above matrix. <table><tr><td>6A</td><td>59</td><td>CB</td><td>BD</td></tr><tr><td>A0</td><td>4E</td><td>48</td><td>12</td></tr><tr><td>30</td><td>9C</td><td>98</td><td>9E</td></tr><tr><td>3D</td><td>F4</td><td>9B</td><td>8B</td></tr></table>	6A	59	CB	BD	A0	4E	48	12	30	9C	98	9E	3D	F4	9B	8B	A	1	<table><tr><td>6A</td><td>59</td><td>CB</td><td>BD</td></tr><tr><td>4E</td><td>48</td><td>12</td><td>A0</td></tr><tr><td>98</td><td>9E</td><td>30</td><td>9C</td></tr><tr><td>8B</td><td>3D</td><td>F4</td><td>9B</td></tr></table>	6A	59	CB	BD	4E	48	12	A0	98	9E	30	9C	8B	3D	F4	9B	<table><tr><td>59</td><td>CB</td><td>BD</td><td>6A</td></tr><tr><td>48</td><td>12</td><td>A0</td><td>4E</td></tr><tr><td>98</td><td>9E</td><td>30</td><td>9C</td></tr><tr><td>8B</td><td>3D</td><td>F4</td><td>9B</td></tr></table>	59	CB	BD	6A	48	12	A0	4E	98	9E	30	9C	8B	3D	F4	9B	<table><tr><td>59</td><td>CB</td><td>BD</td><td>6A</td></tr><tr><td>48</td><td>12</td><td>A0</td><td>4E</td></tr><tr><td>9E</td><td>30</td><td>9B</td><td>9C</td></tr><tr><td>8B</td><td>3D</td><td>F4</td><td>9B</td></tr></table>	59	CB	BD	6A	48	12	A0	4E	9E	30	9B	9C	8B	3D	F4	9B	<table><tr><td>59</td><td>CB</td><td>BD</td><td>6A</td></tr><tr><td>48</td><td>12</td><td>A0</td><td>4E</td></tr><tr><td>9E</td><td>30</td><td>9B</td><td>9C</td></tr><tr><td>8B</td><td>3D</td><td>F4</td><td>9B</td></tr></table>	59	CB	BD	6A	48	12	A0	4E	9E	30	9B	9C	8B	3D	F4	9B
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78	3	Give difference between block cipher and stream cipher.		4																																																																																				
79	3	As we know, AES algorithm has 128-bit key size & 128-bit plain text size. Also, it has 10 rounds & every round has 4 operation. Here, the given data is received after performing first operation of AES algorithm. You have to perform second operation. And then after performing the third operation on second column, show the output for first cell of second column. <table><tr><td>87</td><td>F2</td><td>4D</td><td>97</td></tr><tr><td>EC</td><td>6E</td><td>4C</td><td>90</td></tr><tr><td>4A</td><td>C3</td><td>46</td><td>E7</td></tr><tr><td>8C</td><td>D8</td><td>95</td><td>A6</td></tr></table>	87	F2	4D	97	EC	6E	4C	90	4A	C3	46	E7	8C	D8	95	A6		3																																																																				
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80	3	Give overview of AES including encryption and decryption with neat sketches.		5																																																																																				

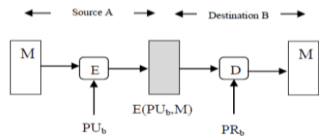
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81	3	Perform AES operation Substitution Bytes from given S-box then also apply shift row on result of substitution byte. $\begin{pmatrix} 00 & 3C & 6E & 47 \\ 1F & 4E & 22 & 74 \\ 0E & 08 & 1B & 31 \\ 54 & 59 & 0B & 1A \end{pmatrix}$ <table><tr><th colspan="16">AES S-box</th></tr><tr><td>00</td><td>01</td><td>02</td><td>03</td><td>04</td><td>05</td><td>06</td><td>07</td><td>08</td><td>09</td><td>0a</td><td>0b</td><td>0c</td><td>0d</td><td>0e</td><td>0f</td></tr><tr><td>00</td><td>63</td><td>7c</td><td>77</td><td>7b</td><td>12</td><td>6b</td><td>6f</td><td>c5</td><td>30</td><td>01</td><td>67</td><td>2b</td><td>fe</td><td>d7</td><td>ab</td></tr><tr><td>10</td><td>ca</td><td>82</td><td>c9</td><td>7d</td><td>fa</td><td>59</td><td>47</td><td>10</td><td>ad</td><td>d4</td><td>a2</td><td>af</td><td>9c</td><td>a4</td><td>72</td></tr><tr><td>20</td><td>b7</td><td>fd</td><td>93</td><td>26</td><td>36</td><td>3f</td><td>f7</td><td>cc</td><td>34</td><td>a5</td><td>e5</td><td>f1</td><td>71</td><td>d8</td><td>31</td></tr><tr><td>30</td><td>04</td><td>c7</td><td>23</td><td>c3</td><td>18</td><td>96</td><td>05</td><td>9a</td><td>07</td><td>12</td><td>80</td><td>e2</td><td>eb</td><td>27</td><td>b2</td></tr><tr><td>40</td><td>09</td><td>83</td><td>2c</td><td>1a</td><td>1b</td><td>6e</td><td>5a</td><td>a0</td><td>52</td><td>3b</td><td>d6</td><td>b3</td><td>29</td><td>e3</td><td>2f</td></tr><tr><td>50</td><td>53</td><td>d1</td><td>00</td><td>ed</td><td>20</td><td>fc</td><td>b1</td><td>5b</td><td>6a</td><td>cb</td><td>be</td><td>39</td><td>4a</td><td>4c</td><td>58</td></tr><tr><td>60</td><td>d0</td><td>ef</td><td>aa</td><td>fb</td><td>43</td><td>4d</td><td>33</td><td>85</td><td>45</td><td>19</td><td>02</td><td>7f</td><td>50</td><td>3c</td><td>9f</td></tr><tr><td>70</td><td>51</td><td>a3</td><td>40</td><td>8f</td><td>92</td><td>9d</td><td>38</td><td>f5</td><td>bc</td><td>b6</td><td>da</td><td>21</td><td>10</td><td>ff</td><td>f3</td></tr><tr><td>80</td><td>cd</td><td>0c</td><td>13</td><td>ec</td><td>5f</td><td>97</td><td>44</td><td>17</td><td>c4</td><td>a7</td><td>7e</td><td>3d</td><td>64</td><td>5d</td><td>19</td></tr><tr><td>90</td><td>60</td><td>81</td><td>4f</td><td>dc</td><td>22</td><td>2a</td><td>90</td><td>88</td><td>46</td><td>ee</td><td>b8</td><td>14</td><td>de</td><td>5e</td><td>0b</td></tr><tr><td>a0</td><td>e0</td><td>32</td><td>3a</td><td>0a</td><td>49</td><td>06</td><td>24</td><td>5c</td><td>c2</td><td>d3</td><td>ac</td><td>62</td><td>91</td><td>95</td><td>e4</td></tr><tr><td>b0</td><td>e7</td><td>c8</td><td>37</td><td>6d</td><td>8d</td><td>d5</td><td>4e</td><td>a9</td><td>6c</td><td>56</td><td>14</td><td>ea</td><td>65</td><td>7a</td><td>ae</td></tr><tr><td>c0</td><td>ba</td><td>78</td><td>25</td><td>2e</td><td>1c</td><td>a6</td><td>b4</td><td>c6</td><td>e8</td><td>dd</td><td>74</td><td>1f</td><td>4b</td><td>bd</td><td>8a</td></tr><tr><td>d0</td><td>70</td><td>3e</td><td>b5</td><td>66</td><td>48</td><td>03</td><td>f6</td><td>0e</td><td>61</td><td>35</td><td>57</td><td>b9</td><td>86</td><td>c1</td><td>1d</td></tr><tr><td>e0</td><td>e1</td><td>f8</td><td>98</td><td>11</td><td>69</td><td>d9</td><td>8e</td><td>94</td><td>9b</td><td>1e</td><td>87</td><td>e9</td><td>ce</td><td>55</td><td>28</td></tr><tr><td>f0</td><td>8c</td><td>a1</td><td>89</td><td>0d</td><td>bf</td><td>e6</td><td>42</td><td>68</td><td>41</td><td>99</td><td>2d</td><td>0f</td><td>b0</td><td>54</td><td>bb</td></tr></table>	AES S-box																00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f	00	63	7c	77	7b	12	6b	6f	c5	30	01	67	2b	fe	d7	ab	10	ca	82	c9	7d	fa	59	47	10	ad	d4	a2	af	9c	a4	72	20	b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	30	04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	40	09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	50	53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	60	d0	ef	aa	fb	43	4d	33	85	45	19	02	7f	50	3c	9f	70	51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	80	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	90	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	a0	e0	32	3a	0a	49	06	24	5c	c2	d3	ac	62	91	95	e4	b0	e7	c8	37	6d	8d	d5	4e	a9	6c	56	14	ea	65	7a	ae	c0	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8a	d0	70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	e0	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	f0	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb		3					
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82	3	Provide a brief overview of the four stages that make up a single round in AES? & explain any one.		2																																																																																																																																																																																																																																																																																																					
83	3	Provide an explanation of the Shift Rows Transformation and mix column in a detail with any Random example.		4																																																																																																																																																																																																																																																																																																					
84	3	-For AES Here, the given data is received after performing substitute bytes operation. -You have to perform operation Sift row then apply mix column on first cell of 3rd column. $\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$		4																																																																																																																																																																																																																																																																																																					
85	3	How do block ciphers differ from stream ciphers?Give name of symmetric block cipher algorithm.		4																																																																																																																																																																																																																																																																																																					
86	3	Discribe the key generation in AES algorithm with diagram.		4																																																																																																																																																																																																																																																																																																					
87	3	What do you understand by Advanced Encryption Standard (AES) ?		3																																																																																																																																																																																																																																																																																																					

UNIT 4 - Public Key Encryption

TOPIC NAME:- Public key Cryptography

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
88	4	The Given figure provides 	A	1	confidentiality	confidentiality and authentication	authentication	None of the mentioned
89	4	The Given figure provides 	C	1	confidentiality	confidentiality and authentication	authentication	None of the mentioned
90	4	Public-key cryptography is also known as ?	A	1	asymmetric cryptography	symmetric cryptography	Both A and B	None of the above
91	4	Which of the following keys are known only to the owner?	C	1	public key	protected key	private key	unique key

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
TOPIC NAME:- RSA Algorithm								
92	4	In the RSA public key cryptosystem, the private and public keys are (e,n) and (d,n) respectively, where $n=p*q$ and p and q are large primes. Let M be an integer such that $0 < M < n$ and $\Phi(n) = (p-1)(q-1)$. Now consider the following equations. I. $M' = Me \bmod n$, $M = (M')d \bmod n$ II. $e d \equiv 1 \bmod n$ III. $e d \equiv 1 \bmod \Phi(n)$ IV. $M' = Me \bmod \Phi(n)$, $M = (M')d \bmod \Phi(n)$ Which of the above equations correctly represent RSA cryptosystem?	B	1	(I) and (II)	(I) and (III)	(II) and (IV)	(III) and (IV)
93	4	Which of the following attacks on RSA algorithm?	D	1	Brute-force attack	Timing attack	Chosen ciphertext attack	All of the above
94	4	What is the primary purpose of the RSA algorithm?	B	1	Symmetric encryption	Public-key encryption	Hashing	Digital signatures
95	4	For $p = 11$ and $q = 17$ and choose $e=7$. Apply RSA algorithm where PT message=88 and thus find the CT.	C	1	23	64	11	54
96	4	For $p = 11$ and $q = 17$ and choose $e=7$. Apply RSA algorithm where Cipher message=11 and thus find the plain text	A	1	88	122	143	111
97	4	In the RSA algorithm, we select 2 random large values 'p' and 'q'. Which of the following is the property of 'p' and 'q'?	C	1	p and q should be divisible by $\Phi(n)$	p and q should be co-prime	p and q should be prime	p/q should give no remainder
98	4	In RSA, $\Phi(n) =$ _____ in terms of p and q.	C	1	$(p)/(q)$	$(p)(q)$	$(p-1)(q-1)$	$(p+1)(q+1)$
99	4	In RSA, we select a value 'e' such that it	C	1	$\gcd(e, \Phi(n))=1$	$1 < e < \Phi(n)$	Both A and B	None of the mentioned
100	4	In the RSA algorithm, we have public key as follows, consider $\gcd(e, \Phi(n))=1$, $1 < e < \Phi(n)$, $d.e \bmod \Phi(n) = 1$	A	1	{e, n}	{d, n}	{d, e}	None of the mentioned
101	4	In the RSA algorithm, we have private key as follows, consider $\gcd(e, \Phi(n))=1$, $1 < e < \Phi(n)$, $d.e \bmod \Phi(n) = 1$	B	1	{e, n}	{d, n}	{d, e}	None of the mentioned
102	4	In the RSA algorithm, value of d such that, consider $\gcd(e, \Phi(n))=1$, $1 < e < \Phi(n)$	A	1	$d.e \bmod \Phi(n) = 1$	$d.e \bmod \Phi(n-1) = 1$	$d.e \bmod \Phi(n+1) = 0$	None of the mentioned
103	4	In the RSA algorithm, plain text message convert in to Cypher text as follows, consider $\gcd(e, \Phi(n))=1$, $1 < e < \Phi(n)$, $d.e \bmod \Phi(n) = 1$	B	1	$C=M^e \bmod n$	$C=M^e \bmod n$	$C=M*d \bmod n$	$C=M^d \bmod n$
104	4	In the RSA algorithm, cypher text message convert in to plain text as follows, consider $\gcd(e, \Phi(n))=1$, $1 < e < \Phi(n)$, $d.e \bmod \Phi(n) = 1$	D	1	$M=C^e \bmod n$	$M=C^e \bmod n$	$M=C*d \bmod n$	$M=C^d \bmod n$
105	4	Perform encryption on the PT using RSA and find the CT. $p = 3$; $q = 11$; $M = 5$	B	1	28	26	18	12
106	4	Perform encryption on the PT using RSA and find the CT. $p = 5$; $q = 11$; $M = 9$	B	1	43	14	26	37
107	4	Perform encryption on the PT using RSA and find the CT. $p = 11$; $q = 13$; $M = 7$		2				
108	4	P and Q are two prime numbers. $P=7$, and $Q=17$. Take $e=5$. If plain text value is 6, then what will be cipher text value according to RSA algorithm?		3				
109	4	Perform encryption and decryption using the RSA algorithm for $p=3$, $q=11$, $e=7$, $M=5$		3				
110	4	In a public key system using RSA, the ciphertext intercepted is $C=10$. which is sent to the user whose public key is $e=5$, $n=35$. What is the plaintext M?		3				
111	4	Perform encryption and decryption using the RSA algorithm for $p=5$, $q=11$, $e=3$, $M=9$.		3				

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INS Practice book (SEM-VI-2026 CE,IT,CSD,AIML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
112	4	P and Q are two prime numbers. P=3, and Q=7. Take $e=5$. If plain text value is 5, then what will be cipher text value according to RSA algorithm? Alos find private key and public key.		4				
113	4	P and Q are two prime numbers. P=3, and Q=11. Take $e=3$. If plain text value is 5, then what will be cipher text value according to RSA algorithm? Alos find private key and public key.		4				
114	4	P and Q are two prime numbers. P=5, and Q=11. Take $e=3$. If plain text value is 9, then what will be cipher text value according to RSA algorithm? Alos find private key and public key.		4				
115	4	P and Q are two prime numbers. P=11, and Q=13. Take $e=11$. If plain text value is 8, then what will be cipher text value according to RSA algorithm? Alos find private key and public key.		4				
116	4	P and Q are two prime numbers. P=7, and Q=11. Take $e=17$. If plain text value is 8, then what will be cipher text value according to RSA algorithm? Alos find private key and public key.		4				
117	4	P and Q are two prime numbers. P=3, and Q=11. Take $e=7$. If plain text value is 5, then what will be cipher text value according to RSA algorithm? Alos find private key and public key.		4				
TOPIC NAME:- Diffie Hellman Key Exchange Algorithm								
118	4	What is the fundamental idea behind the Diffie-Hellman key exchange?	D	1	Sharing of public keys	Mutual authentication	Digital signatures	Securely exchanging secret keys over an insecure channel
119	4	In the Diffie-Hellman key exchange algorithm, which of the following is NOT exchanged between the parties?	B	1	Public keys	Private keys	Shared secret	Prime modulus and generator
120	4	How many bytes of the secret key is generated using Diffie-Hellman encryption/decryption scheme?	A	1	256	871	1024	962
121	4	Diffie-Hellman algorithm is widely known as	C	1	key exchange algorithm	key agreement algorithm	both a and b	only a
122	4	In the Diffie-Hellman key exchange algorithm consider prime number q, and suppose user A and B exchange a key then Pr Key of user A is always	A	1	$X_a < q$	$X_a > q$	$X_a < q-2$	$X_a > q-2$
123	4	In the Diffie-Hellman key exchange algorithm consider prime number q, and suppose user A and B exchange a key then Pr Key of user B is always	A	1	$X_b < q$	$X_b > q$	$X_b < q-2$	$X_b > q-2$
124	4	Consider a Diffie-Hellman scheme with a common prime $q=5$ and a primitive root $\alpha=3$. (i) Show that 3 is a primitive root of 5.		2				
125	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) Show that 2 is a primitive root of 11.		2				
126	4	Consider a Diffie-Hellman scheme with a common prime $q=5$ and a primitive root $\alpha=4$. check that 4 is a primitive root of 11. or Not?		2				
127	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) Show that 2 is a primitive root of 11. (ii) If user A has private key $X_A=3$, what is A's public key Y_A ?		2				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
128	4	Consider a Diffie-Hellman scheme with a common prime $q=23$ and a primitive root $\alpha=5$. (i) If user A has private key $X_A=3$, what is A's public key Y_A ? (i) If user B has private key $X_B=4$, what is B's public key Y_B ?		2				
129	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) Show that 2 is a primitive root of 11. (ii) If user B has private key $X_B=7$, what is B's public key Y_A ?		2				
130	4	Consider a Diffie-Hellman scheme with a common prime $q=23$ and a primitive root $\alpha=5$. (i) If user A has private key $X_A=3$, what is A's public key Y_A ? And A's secret key S_A ? (i) If user B has private key $X_B=4$, what is B's public key Y_B ? And B's secret key S_B ?		4				
131	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) If user A has private key $X_A=3$, what is A's public key Y_A ? And A's secret key S_A ? (i) If user B has private key $X_B=7$, what is B's public key Y_B ? And B's secret key S_B ?		4				
132	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) If user A has private key $X_A=8$, what is A's public key Y_A ? And A's secret key S_A ? (i) If user B has private key $X_B=4$, what is B's public key Y_B ? And B's secret key S_B ?		4				
133	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) Show that 2 is a primitive root of 11. (ii) If user A has public key $Y_A=9$, what is A's private key X_A ? (iii) If user B has public key $Y_B=3$, what is A's private key X_B & what is the shared secret key K ?		4				
134	4	Consider a Diffie-Hellman scheme with a common prime $q=11$ and a primitive root $\alpha=2$. (i) Show that 2 is a primitive root of 11. (ii) If user A has private key $X_A=9$, what is A's public key Y_A ? (iii) If user B has private key $X_B=3$, what is B's public key Y_B & what is the shared secret key of users.		4				

UNIT 5 - Message Digest

TOPIC NAME:- Hash function

135	5	What is MAC?	B	1	Message Activation Code	Message Authentication Code	Massive Authentication Code	Massive Activation Code
136	5	Typical hash functions take inputs of variable lengths to return outputs of a	A	1	fixed length	variable length	double length	none of these
137	5	Hash function is also known as	C	1	Expansion function	composite function	Compression function	Excessive function
138	5	Message digests should not reveal anything about the _____.	D	1	key	password	bit size	original message

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
139	5	Message Authentication Code is also known as	A	1	hash code	key code	private code	all of above
140	5	How many methods are there to provide C.I.A?	C	1	2	4	6	8
141	5	What is C.I.A?	A	1	Confidentiality, Integrity, Authentication	Confidentiality, Integrity, Activation	Compression, Integrity, Authentication	Compression, Integrity, Activation
142	5	What is hash function? What is th use of hash function?		3				
143	5	Which are the different methods to provide authentiction using hash function? Write in detail.		6				
144	5	Draw the diagram which provide authentication and confidentiality both using hash function with symmetric key.		3				
145	5	Draw the diagram which provide only authentication and no confidentiality using hash function with symmetric key.		3				
146	5	Draw the diagram which provide only authentication and no confidentiality using hash function with asymmetric key.		3				
TOPIC NAME:- MD5								
147	5	MD5 algorithm was developed in which year?	D	1	1995	1994	1992	1991
148	5	MD5 algorithm produces _____ bit message digest.	D	1	16	64	32	128
149	5	MD5 algorithm was developed by,	B	1	Ronald Nephew	Ronald Rivest	Ronald Newton	none of these
150	5	The first step in MD5 is to add _____ bits to the original message.	C	1	chaining	initialize	padding	appending
151	5	From the following which step aim is to make the length of the original message equal to a value, which is 64 bits less than an exact multiple of 512.	C	1	chaining	initialize	padding	appending
152	5	How many chaining variables are there in MD5 algorithm?	B	1	5	4	6	8
153	5	The padding consists of a single 1-bit, followed by as many _____, as required.	C	1	2-bits	4-bits	0-bits	16-bits
154	5	How many number of rounds are there in MD5 algorithm?	C	1	5	3	4	6
155	5	What is the block size in MD5 algorithm?	A	1	512 bits	128 bits	32 bits	64 bits
156	5	Which is the second step of MD5 algorithm?	D	1	chaining	initialize variable	padding	append length
157	5	How MD5 algorithm works, describe stepwise with figure.		5				
158	5	Draw MD5 algorithm operation diagram.		2				
159	5	How padding operation done in MD5 algorithm? Also mention importance of padding operation.		4				
160	5	How four number of rounds are performed in MD5 algorithm?		5				
161	5	What is initializing of chaining variable in MD5?		2				
TOPIC NAME:- SHA512								
162	5	What is SHA?	C	1	Single Hash Algorithm	Secure Host Algorithm	Secure Hash Algorithm	Single Host Algorithm
163	5	SHA was developed by,	A	1	NIST, NSA	NSTI, NSA	FPIS, NSTI	FPIS, NSA
164	5	SHA algorithm was developed in which year?	D	1	1995	1994	1992	1993
165	5	SHA512 algorithm produces _____ bit message digest.	A	1	512	64	32	128
166	5	What is the block size in SHA512 algorithm?	C	1	512 bits	128 bits	1024 bits	64 bits
167	5	The SHA512 algorithm takes a message of length _____ bits.	B	1	2^{32}	2^{128}	2^{64}	2^{512}
168	5	How many chaining variables are there in SHA512 algorithm?	D	1	5	4	6	8
169	5	How many number of rounds are there in SHA512 algorithm?	C	1	50	25	80	60
170	5	Give comparision of MD5 and SHA512 algorithm.		5				
171	5	Explain SHA512 algorithm including all steps in detail.		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
172	5	Draw SHA512 algorithm diagram.		4				
173	5	What is initializing of chaining variable in SHA512?		3				
174	5	Compare MD5 and SHA512 based on following parameter: (a) message digest size, (b) message size, (c) block size (d) steps in algorithm.		4				
UNIT 6 - Message Integrity and Message Authentication								
TOPIC NAME:- Message Authentication and its requirements and security								
175	6	When a message has not been altered, it is called as,	B	1	hash code	Message integrity	Message code	all of above
176	6	The most common approach of message integrity is to use a _____ that combines all the bytes in the message with a secret key and produces a message digest that is difficult to reverse.	C	1	Expansion function	composite function	hash function	none of these
177	6	The most common approach of message integrity is to use a hash function that combines all the bytes in the message with a secret key and produces a _____ that is difficult to reverse.	A	1	message digest	private key	public key	code
178	6	In cryptography, a message authentication code (MAC), sometimes known as an	D	1	authentication key	authentication code	authentication layer	authentication tag
179	6	Using MAC finction, how many methods are there to achieve authentication?	D	1	2	4	5	3
180	6	Another name of Message Authentication Codes is,	A	1	Cryptographic checksum	Cryptographic codesum	Cryptographic codebreak	Cryptographic checkbreak
181	6	Releasing the content of the message to someone who does not have all appropriate cryptographic key is called as,	B	1	Modification of content	Revelation	Source refusal	Modification of timing
182	6	When the source denies being the originator of a message is called as,	C	1	Modification of content	Revelation	Source refusal	Modification of timing
183	6	Which authentication requirement includes insertion, deletion and reordering of messages?	B	1	Modification of content	Modification in sequence	Revelation	Modification of timing
184	6	When the receiver of the message denies the reception, is called as	C	1	Modification of content	Revelation	Defination refusal	Modification of timing
185	6	Which authentication requirement includes replay and delay of messages sent between different parties?	D	1	Modification of content	Modification in sequence	Revelation	Modification in timing
186	6	What is messgae integrity?		2				
187	6	What is message authentication?		2				
188	6	Which are the two functaionality levels of message authentication?. Discuss that in brief.		3				
189	6	How authentication can be grouped into three classes?		3				
190	6	Which are the authentiction requirements? List and explain all requirements.		7				
191	6	In authentication requirements, What is revelation and analysis of traffic?		3				
192	6	For authentication requirements describe the following terms: (a) modification in the content (b) modification in the sequence (c) modification in the timing.		3				
193	6	What is source refusal in authentication requirement?		3				
194	6	What is defination refusal in authentication requirement?		3				
195	6	Whatis authentication requirements? Explainn with example.		4				
TOPIC NAME:- Message Authentication Code, HMAC								
196	6	What is HMAC?	B	1	Hash based Message Activation Code	Hash based Message Authentication Code	Hash based Massive Authentication Code	Hash based Massive Activation Code

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
197	6	HMAC has been chosen as a mandatory security implementation for the _____ security and is also used in the _____ protocol, widely used on the internet.	A	1	IP, SSL	IP, HASH	MAC, SSL	MD, SSL
198	6	What is the value of ipad in the HMAC structure?	A	1	00110110	100001010	11110000	10101010
199	6	In HMAC algorithm, make the length of shared symmetric key equal to	C	1	no. of bits in different block	no. of codes	no. of bits in original msg block	line of code
200	6	What is the value of opad in the HMAC structure?	A	1	01011010	100001010	11110000	10101010
201	6	In HMAC algorithm, if the length of shared symmetric key is greater than no. of bits in each block, then _____ is apply to compress the key.	C	1	Expansion function	composite function	hash function	code function
202	6	In HMAC algorithm, the sender and receiver only know about the	A	1	symmetric key	asymmetric key	hidden key	final key
203	6	One of the disadvantages of HMAC is, it can not be used if the number of receivers is greater than	B	1	2	1	3	4
204	6	HMAC uses the shared _____ to encrypt the message digest, which produces the output MAC.	A	1	symmetric key	asymmetric key	hidden key	final key
205	6	What is Message authentication code?		2				
206	6	To achieve authentication using MAC, which three methods are available?		3				
207	6	Draw the diagram of MAC for authentication only.		3				
208	6	Draw the diagram of MAC for authentication and confidentiality both, and authentication tied to plain text.		3				
209	6	Draw the diagram of MAC for authentication and confidentiality both, and authentication tied to cipher text.		3				
210	6	What is Hash based Message authentication code?		2				
211	6	How HMAC algorithm works? Describe in brief.		4				
212	6	If the length of the shared symmetric key used in HMAC (k) is less than the number of bits in each block (b), then how we can make k=b.		3				
213	6	If the length of the shared symmetric key used in HMAC (k) is greater than the number of bits in each block (b), then how we can make k=b.		3				
214	6	Draw the diagram showing complete HMAC operation.		5				

UNIT 07- DIGITAL SIGNATURE

TOPIC: Digital Signature: Properties, Requirements, & Security

215	7	What is a digital signature?	B	1	A scanned image of a handwritten signature	A cryptographic technique to verify the authenticity and integrity of a digital message or documents	A biometric authentication method	A watermark applied to electronic images
216	7	Which of the following is a primary purpose of digital signatures	C	1	Ensuring data confidentiality	Providing proof of identity	Verifying the integrity and authenticity of digital data	Encrypting digital communication channels
217	7	Which of the following is a property of a digital signature?	D	1	It must verify the author and the date and time of the signature	It must to authenticate the contents at the time of the signature	It must be verifiable by third parties, to resolve disputes	All of the above

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
218	7	Which of the following is a requirement for a digital signature?	D	1	Must be bit pattern depending on the message being signed	Computationally easy to produce a signature	Computationally easy to recognize and verify & verify the signature	All of the above
219	7	How many types or approaches are proposed for the digital signature function?	C	1	3	4	2	5
220	7	In an arbitrated digital signature scheme, who is responsible for generating the digital signature?	C	1	The sender	The recipient	A trusted third party	A public key infrastructure (PKI)
221	7	In a direct digital signature scheme, who is responsible for generating the digital signature?	A	1	The sender	The recipient	A trusted third party	A public key infrastructure (PKI)
222	7	What is the need of digital signature? Explain with example.		3				
223	7	What are the properties of Digital Signature?		3				
224	7	List out the security services provided by digital signature?		3				
225	7	Compare Direct Digital Signature with Arbitrated Digital Signature.		4				
226	7	What are the requirements of digital signature?		5				
227	7	Enlist the approaches and types has been proposed for the digital signature function and elaborate any one approach or type to digital signatures.		5				
TOPIC: Digital SignatureAlgorithm (DSA) or Digital Signature Standarad (DSS)								
228	7	DSA stands for?	C	1	Data Structure & Algorithms	Digital Signal Algorithm	Digital Signature Algorithm	Digital Standarad Algorithm
229	7	Digital Signature Algorithm is also known as?	B	1	Digital Structure Standard	Digital Signature Standard (DSS)	Digital Structure Algorithm	Data Structure Solver
230	7	Digital Signature Algorithm (DSA) proposed in which year?	C	1	1992	1990	1991	1993
231	7	Which organization developed the DSA and standardized in which year?	D	1	NIST, 1991	IETF, 1990	ISO, 1992	NIST, 1994
232	7	Which security service is not provided by digital signature?	D	1	Integrity	Authentication	Non-repudiation	Access control
233	7	Which algorithm is commonly used for creating digital signatures?	C	1	AES (Advanced Encryption Standard)	SHA (Secure Hash Algorithm)	RSA (Rivest-Shamir-Adleman)	HMAC (Keyed-Hashing for Message Authentication)
234	7	What cryptographic operation is performed to create a digital signature?	B	1	Encryption	Hashing	Decryption	Compression
235	7	Vishal digitally signs a message and sends it to Hardik. Verification of the signature by Hardik requires?	A	1	Vishal's public key	Hardik's public key	Hardik's private key	Vishal's private key
236	7	Which one of the following is an advantage of the digital signature algorithm?	D	1	Highly robust	Less storage requirements	Better speed in computations.	All of the above
237	7	Digital signature cannot provide _____ for the message?	C	1	Integrity	Authentication	Confidentiality	Hashing
238	7	In a digital signature algorithm, a prime number p is selected with a length of how many bits?	B	1	128 and 512 bits	512 and 1024 bits	64 and 128 bits	32 and 64 bits
239	7	Digital signature standard cannot be used for?	C	1	Decryption	Authorization	Encryption or key excahnge	All of the above
240	7	In digital signatures, which key is used for signature verification?	A	1	Public key	Private key	Session key	Symmetric key
241	7	For a digital signature, what type of relationship is there between signature and a message?	D	1	One to Many	Many to One	Many to Mamy	One to One
242	7	In digital signature algorithm, verification process done by which key?	A	1	Public key	Private key	Session key	Secret key

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
243	7	In the digital signature algorithm, what type of key pair is used?	B	1	Symmetric key pair	Public-private key pair	Hash key pair	Encryption key pair
244	7	Explain digital signature algorithm (DSA).		5				
245	7	What is the principle of digital signature algorithm (DSA). How a user can create a signature using DSA? Explain the signing and verifying function in DSA.		5				
246	7	Define digital signature? Explain its use with the help of example.		5				
247	7	Given data for Digital Signature Algorithm (DSA): $H(m) = 3$, $P = 7$, $h = 2$ and $k = 2$ Calculate the above problem for the signing and verifying function in DSA.		5				
248	7	Given data for Digital Signature Algorithm (DSA): $H(m) = 3$, $P = 23$, $h = 3$, $q = 11$, $x = 7$, $g = 9$ and $k = 5$ Calculate the above problem for the signing and verifying function in DSA.		5				
249	7	Given data for Digital Signature Algorithm (DSA): $P = 23$, $q = 11$, $g = 4$, $x = 2$, $k = 7$, $H(m) = 8$ Calculate the above problem for the signing and verifying function in DSA.		5				
250	7	Given data for Digital Signature Algorithm (DSA): $P = 23$, $q = 11$, $g = 4$, $x = 3$, $k = 6$, $H(m) = 9$ Calculate the above problem for the signing and verifying function in DSA.		5				
251	7	Given data for Digital Signature Algorithm (DSA): $P = 23$, $q = 11$, $g = 2$, $x = 6$, $k = 3$, $H(m) = 9$ Calculate the above problem for the signing and verifying function in DSA.		5				
252	7	Given data for Digital Signature Algorithm (DSA): $P = 23$, $q = 11$, $g = 4$, $x = 3$, $k = 5$, $H(m) = 4$ and $h = 2$ Calculate the above problem for the signing and verifying function in DSA.		5				
TOPIC: Digital Signature Scheme [ElGamal Digital Signature Scheme]								
253	7	Which digital signature scheme is based on Diffie-Hellman key exchange?	A	1	Elgamal digital signature scheme	RSA (Rivest-Shamir-Adleman digital signature scheme)	Qualified electronic signature scheme	SHA (Secure Hash Algorithm) digital signature scheme
254	7	Which key is used by the recipient to verify the digital signature in the ElGamal digital signature scheme?	A	1	Public key	Private key	Session key	Symmetric key
255	7	What type of key pair is used in the ElGamal Digital Signature algorithm?	B	1	Symmetric key pair	Public-private key pair	Hash key pair	Encryption key pair
256	7	What key is used to verify the digital signature in ElGamal Digital Signature?	A	1	Public key	Private key	Symmetric key	Session key
257	7	ElGamal digital signature scheme is one of the first digital signature scheme based on an _____ as prime.	B	1	Exponential Function	Arithmetic Modulo	Finite field function	Elliptic curve function
258	7	For Elgamal digital signature scheme alongwith authentication, if we want confidentiality of message the encryption can be done by using which key?	A	1	Public key	Private key	Secret key	Symmetric key
259	7	Which of the following is NOT a step in ElGamal signature algorithm?	C	1	Key Generation	Signature Verification	Hashing	Signature Generation
260	7	Explain Elgamal digital signature scheme in detail.		5				
261	7	Calculate the example for signature and verifying using ElGamal Signature having the following data: $q = 23$, $\alpha = 5$, Private key = 6, $H(m) = 7$ and $k = 5$		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
262	7	Consider ElGamal signature in $q = 19$ with generator 10 and given private key is 16 and hash value $m = 14$. Calculate the example for signature and verifying.		5				
263	7	Consider ElGamal signature in $q = 17$ with generator 8 and given private key is 9 and hash value $m = 12$. Calculate the example for signature and verifying.		5				
264	7	Calculate the example for signature and verifying using ElGamal Signature having the following data: $q = 13$, $\alpha = 2$, Private key = 3, $m = 11$ and $k = 5$		5				
265	7	Calculate the example for signature and verifying using ElGamal Signature having the following data: $q = 23$, $\alpha = 5$, Private key = 6, $H(m) = 10$ and $k = 3$		5				
UNIT 8 - Key Exchange								
TOPIC NAME:- Key management and distribution, X.509 Digital Certificate, Public Key Infrastructure (PKI), Kerberos								
266	8	Which of the following is true about Public Key Infrastructure?	A	1	PKI is a combination of digital certificates, public-key cryptography, and certificate authorities that provide enterprise wide security.	PKI uses two-way symmetric key encryption with digital certificates, and Certificate Authority.	PKI uses private and public keys but does not use digital certificates.	PKI uses CHAP authentication.
267	8	The subject unique identifier of the X.509 certificates was added in which version?	B	1	1	2	3	4
268	8	Which of the following is not an element/field of the X.509 certificates?	B	1	Issuer Name	Serial Modifier	Issuer unique Identifier	Signature
269	8	Which one of the following is not a public key distribution means?	B	1	Public-Key Certificates	Hashing Certificates	Publicly available directories	Public-Key authority
270	8	Which of the following public key distribution systems is most secure?	A	1	Public-Key Certificates	Public announcements	Publicly available directories	Public-Key authority
271	8	Which systems use a timestamp?i) Public-Key Certificatesii) Public announcementsiii)	C	1	i) and ii)	iii) and iv)	i) and iv)	iv) only
272	8	Which of these systems use timestamps as an expiration date?	A	1	Public-Key Certificates	Public announcements	Publicly available directories	Public-Key authority
273	8	X.509 certificate recommends which cryptographic algorithm?	A	1	RSA	DES	AES	Rabin
274	8	How many functions are involved in the PKIX architectural model?	D	1	3	5	6	7
275	8	Extensions were added in which version?	C	1	1	2	3	4
276	8	It is desirable to revoke a certificate before it expires because	D	1	the user is no longer certified by this CA	the CA's certificate is assumed to be compromised	the user's private key is assumed to be compromised	all of the mentioned
277	8	CRL stands for	C	1	Cipher Reusable List	Certificate Revocation Language	Certificate Revocation List	Certificate Resolution Language
278	8	Publicly Available directory is more secure than which other system?	B	1	Public-Key Certificates	Public announcements	Public-Key authority	None of the mentioned
279	8	MongoDB supports _____ certificate authentication for use with a secure TLS/SSL connection.	B	1	x.590	x.509	x.950	All of the mentioned
280	8	For a client-server authentication, the client requests from the KDC a _____ for access to a specific asset.	A	1	ticket	local	token	user

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
281	8	The standard used in digital certificates that defines its structure, fields, and values is _____.	A	1	X.509	Kerberos	Cryptography	PKI
282	8	Key management is a practice that requires:	A	1	Choosing a key that is extremely random and the algorithm should use the full range of the key-space	Labeling keys so that they are not lost or stolen	Returning the key to the CA after it has completed its lifetime	At least two senior officers of the company to issue and maintain a record of the keys
283	8	Which port does Kerberos use?	B	1	80	88	25	23
284	8	Kerberos was developed at _____.	A	1	MIT	Oxford	Stanford	None of them
285	8	Kerberos takes into account _____.	D	1	Client	KDC	TGS	All of them
286	8	_____ public key infrastructure (PKI) standard identifies the format of public key certificates.	B	1	X.500	X.509	X.590	X.540
287	8	USENET is related to which of the following Public Key distribution schemes?	B	1	Public-Key Certificates	Public announcements	Publicly available directories	Public-Key authority
288	8	What are the various elements of public key infrastructure?	D	1	Certificate authority	Registration authority	Certificate database	All of them
289	8	Example of an Authentication Token is	B	1	Key fob	Smart card	Pin	None of these
290	8	Which of the following are the ways through which virus spreads?	D	1	Floppy Disk	CD	Email attachments	All of these
291	8	For confidentiality, data to be sent is	C	1	Corrected	Decrypted	Encrypted	Both (a) and (b)
292	8	The issuer unique identifier of the X.509 certificates was added in which version?	B	1	1	2	3	4
293	8	What is the name of the issuer of public key infrastructure?	A	1	Certificate authorities	Registration authority	Client	Verification authority
294	8	Public key encryption/decryption is not preferred because	D	1	it is slow	it is hardware/software intensive	it has a high computational load	all of the mentioned
295	8	Which of the following factors must be considered when implementing Kerberos authentication? Select one:	A	1	Kerberos requires a centrally managed database of all user and resource passwords.	Kerberos tickets can be spoofed using replay attacks to network resources.	Kerberos can be susceptible to man in the middle attacks to gain unauthorized access.	Kerberos uses clear text passwords.
296	8	List out the general categories of schemes for the distribution of public keys and Elaborate two general categories of schemes for the distribution of public keys with proper illustrations.		3				
297	8	List out four general categories of schemes for the distribution of public keys and Elaborate various methods for the distribution of public keys with proper illustrations.		5				
298	8	What do you mean by key distribution? Explain public- key authority and public-key certificate method for key distribution with proper illustration.		3				
299	8	Define and discuss various components of public key infrastructure (PKI).		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
300	8	Discuss X.509 Certificates.		5				
301	8	Draw X.509 certificate format. Enlist the reasons to revoke the certificate before expiry.		3				
302	8	Why is PKI used?		3				
303	8	How Does PKI Work? What Is a Certificate Authority (CA) and Registration authority (RA)?		3				
304	8	Explain Kerberos in detail.		5				
305	8	What is Kerberos? How Kerberos authenticates the users for authorized service access?		5				
306	8	Explain through diagram, how Kerberos can communication with other administrative domains for providing trusted services to the clients.		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
307	8	Enlist requirements of Kerberos. Enlist and explain roles of various servers are used in Kerberos.		3				
308	8	Explain through diagram, how Kerberos can communicate with HTTP server for providing trusted services to you. (Write your name in place of user)		3				
309	8	What Is the difference between Public key Cryptography and Public key Infrastructure.		2				
310	8	Explain public announcement and public key directories method for key distribution with proper illustration.		2				
UNIT 9- Network Security								
TOPIC NAME:- PGP,Secure Socket Layer, IP Security								
311	9	Who is the creator of PGP?	A	1	Phil Zimmerman	Tim Berners-Lee	Linus Torvalds	Steve Jobs
312	9	PGP is primarily used for:	B	1	Secure File Transfer	Email Encryption	Web Browsing	Network Firewalling
313	9	Which cryptographic algorithms are commonly used in PGP?	A	1	RSA and AES	DES and SHA	MD5 and Triple-DES	Diffie-Hellman and Blowfish
314	9	In PGP, what is the purpose of a "revocation certificate"?	C	1	To decrypt encrypted messages	To authenticate the sender of a message	To revoke a compromised or lost private key	To establish secure connections between servers
315	9	Which component of a PGP key pair is kept secret by the owner?	B	1	Public Key	Private Key	Encryption Key	Digital Signature
316	9	In PGP, what is the purpose of the "Keyserver"?	B	1	To generate PGP key pairs	To distribute public keys	To decrypt encrypted messages	To perform digital signatures
317	9	Which version of PGP introduced support for the OpenPGP standard?	B	1	PGP 2.0	PGP 5.0	PGP 8.0	PGP 10.0
318	9	In PGP encryption, which key is used for encrypting the message?	C	1	Sender's Public Key	Sender's Private Key	Recipient's Public Key	Recipient's Private Key
319	9	How does PGP ensure the confidentiality of messages?	A	1	By using symmetric-key encryption	By encrypting messages with the sender's private key	By using a secure channel for communication	By hashing the message content
320	9	Which cryptographic algorithm is commonly used for generating PGP key pairs?	C	1	MD5	SHA-256	RSA	AES
321	9	List the techniques which are used in PGP and draw the work flow of it.		2				
322	9	how can PGP provide Authentication and Digital signature? Explain it with diagram.		2				
323	9	How can PGP obtain only confidentiality? Explain with neat diagram.		2				
324	9	Can PGP provide confidentiality and authentication? Explain with diagram.		2				
325	9	Illustrate PGP with it's working operation.		2				
326	9	Illustrate PGP with any one case by using neat diagram.		3				
327	9	Enlist the various security protocols used in SSL and illustrate Record protocol.		2				
328	9	What are the four phases of SSL protocol?		2				
329	9	What is SSL Alert protocol ? Elobrate alerts with description.		3				
330	9	Enlist the type of SSL protocol? Explain any one of them.		3				
331	9	What does SSL provide in terms of security features?	B	1	Confidentiality and authentication	Confidentiality, integrity, and authentication	Integrity and authorization	None of this

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
332	9	Which cryptographic algorithm is commonly used in SSL for key exchange and encryption?	A	1	RSA	MD5	DES	SHA-256
333	9	Which layer of the OSI model does SSL operate at?	C	1	Application layer	Network layer	Transport layer	Data link layer

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
334	9	In SSL communication, what does the 'handshake' phase accomplish?	B	1	Encrypting data	Exchanging cryptographic keys	Authenticating the server	Initiating the connection
335	9	Which organization is responsible for the development and standardization of SSL protocols?	C	1	IEEE	ISO	IETF	LJU
336	9	Which cryptographic techniques are used by the SSL Record Protocol for secure communication?	D	1	Asymmetric encryption	Symmetric encryption	Hashing	All
337	9	Which of the following messages is NOT part of the SSL handshake protocol?	D	1	ClientHello	ServerHello	Certificate Request	DataTransfer
338	9	Which message in the SSL handshake protocol confirms that both client and server have completed the handshake process?	B	1	ServerHelloDone	Finished	Certificate Verify	Change CipherSpec
339	9	Which type of alert message is sent when a fatal error occurs during SSL communication, indicating that the connection must be terminated?	C	1	Warning alert	Close notify alert	Fatal alert	HelloRequest alert
340	9	Which alert message is commonly used to indicate that the SSL connection is being terminated ?	B	1	Fatal alert	Close notify alert	Warning alert	Handshake failure alert
341	9	What is the purpose of a warning alert in the SSL protocol?	A	1	To indicate a non-fatal issue that may affect the connection	To authenticate the server to the client	To request renegotiation of cryptographic parameters	To terminate the connection immediately due to a severe error
342	9	In the SSL Alert Protocol, what does the description "bad_record_mac" typically indicate?	A	1	A cryptographic integrity check failure on received data	A failure to authenticate the server	A mismatch in protocol versions	A cryptographic algorithm negotiation failure
343	9	Which of the following IPsec modes encrypts both the IP header and the payload?	A	1	Tunnel mode	Transport mode	ESP mode	AH mode
344	9	In IPsec, what is the purpose of the Authentication Header (AH) protocol?	B	1	To provide encryption for payload data	To provide data integrity and authentication	To negotiate encryption algorithms	To encapsulate the entire IP packet for secure transmission
345	9	Which mode of IPsec provides end-to-end encryption between hosts?	C	1	Tunnel mode	VPN mode	Transport mode	Network Address Translation mode
346	9	What is the primary purpose of IPsec?	B	1	To encrypt data at the application layer	To secure communications at the network layer	To authenticate users during login	To manage domain name resolution
347	9	Which mode of ESP is commonly used to create a virtual private network (VPN) between two gateways?	A	1	Tunnel mode	Transport mode	Payload mode	Route mode
348	9	What is the size of Security Parameters Index (SPI) field in the Authentication Header in Ipsec?	B	1	16 bits	32 bits	64 bits	128 bits
349	9	Which type of certificate is commonly used to authenticate servers in SSL?	D	1	Private Key Certificate	SSL Certificate	Public Key Certificate	Digital Certificate
350	9	Which SSL Alert message indicates that the certificate chain received from the server is not trusted?	D	1	Certificate revoked	Certificate expired	Certificate unknown	Bad certificate

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
351	9	What is the primary purpose of the SSL Alert protocol?	C	1	To establish secure connections between client and server	To negotiate cryptographic algorithms for encryption	To communicate security-related alerts and errors	To exchange public keys for authentication
352	9	What is the primary purpose of the Change Cipher Spec Protocol in SSL?	C	1	To negotiate cryptographic algorithms for encryption	To authenticate the server to the client	To signal the transition to encrypted communication	To exchange public keys for encryption
353	9	Which phase of the SSL handshake involves the exchange of cryptographic parameters and keys ?	B	1	Authentication	Key Exchange	Cipher Specification	Handshake Completion
354	9	What does the SSL Record Protocol use to protect against data tampering?	C	1	Public key cryptography	Digital signatures	MAC (Message Authentication Code)	Symmetric encryption
355	9	Which key management protocol is used to exchange cryptographic keys in IPsec?	A	1	IKE	Kerberos	SSH	SSL
356	9	What is the use of IPsec? Give atleast one protocol with neat diagram.		2				
357	9	List the Ipvsec protocols and Applications.		2				
358	9	Explain Authentication Header with neat diagram.		2				
359	9	What is ESP? Explain it with neat diagram.		2				
360	9	State two mode of operation and explain with diagram.		3				

UNIT 10- Network Designing and Monitoring

TOPIC NAME:-VLAN,VPN

361	10	What does VLAN stand for?	B	1	Virtual Logical Area Network	Virtual Local Area Network	Virtual Linear Area Network	Virtual Local Access Network
362	10	VLANs operate at which layer of the OSI model?	B	1	Physical Layer	Data Link Layer	Network Layer	Transport Layer
363	10	Which VLAN type is designed specifically for user-generated data?	B	1	Management VLAN	Data VLAN	Voice VLAN	Default VLAN
364	10	Which VLAN type is referred to when all ports on a device belong to it by default when the switch is on?	C	1	Management VLAN	Data VLAN	Default VLAN	Native VLAN
365	10	What is the primary purpose of a VLAN?	B	1	To connect multiple physical LAN segments	To group one or more LANs for communication based on logical connections	To increase the number of network devices	To physically connect devices in a network
366	10	How does a VPN work?	B	1	By routing internet traffic directly to your ISP	By routing internet traffic through a specially configured remote server	By routing internet traffic through a local switch	By using a combination of Wi-Fi and mobile networks
367	10	What is the main benefit of using VPNs for remote access?	C	1	Increased network speed	Reduced security risks	Securely connecting remote users to the corporate network	Access to restricted websites
368	10	What is one of the main benefits of using OpenVPN?	C	1	It uses outdated encryption protocols	It is not supported on many operating systems	It is highly configurable and secure	It does not support SSL/TLS encryption
369	10	Which VPN technology is specifically designed to connect remote branch offices in a secure manner?	B	1	Remote Access VPN	Site-to-Site VPN	Layer 2 Tunneling Protocol (L2TP)	OpenVPN
370	10	In a VPN tunnel, what does data encryption help to achieve?	C	1	Increase data transfer speed	Reduce network latency	Protect data confidentiality	Improve network reliability